

Principles of Economics

Class 5: The Firm and its Customers

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Plan for today's class

Introduction

The firm's problem

What is feasible: demand

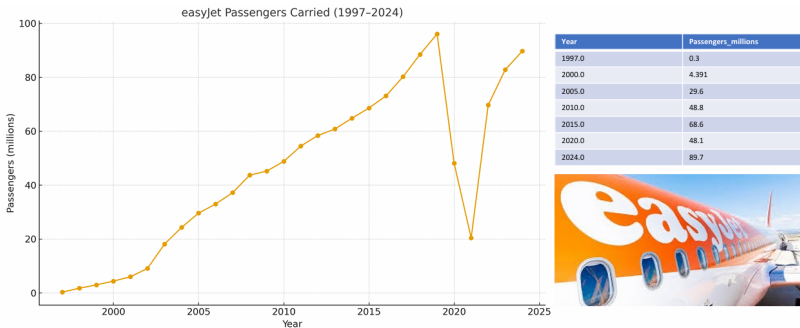
Costs

What is desirable: isoprofit

Profit maximisation

1. Introduction

- How can a firm become a market leader competing on low prices?



- Keys to a successful pricing policy?

2. The firm's problem

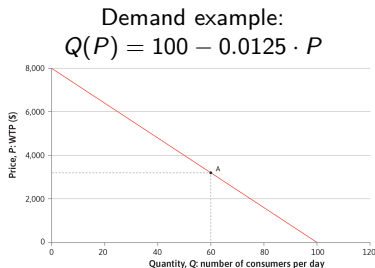
- ▶ **Profitability:** a common measure of a firm's success:
 - ▶ We **model how firms choose output and price**.
- ▶ Simplified version of the **firm's problem**: firms maximise profits:
$$\pi(P, Q) = R(P, Q) - C(Q)$$
 - ▶ Where:
 - ▶ R : **total revenue** from selling Q units at price P ;
 - ▶ C : **total cost**, depends on output.
- ▶ As always: what is feasible, then what is desirable.

3. What is feasible: demand

What is feasible

Demand curve

- ▶ If the firm sets its price, what stops it from setting the highest one?
- ▶ Consumers:
 - ▶ too high a price, few buyers.
- ▶ This is summarised by the **demand curve**.
 - ▶ **Maximum quantity demanded at each price:**
 $Q(P) = a - b \cdot P$
 - ▶ Usually, the **slope is negative**: $\uparrow P \Rightarrow \downarrow Q$
- ▶ **Estimated** with surveys or econometrics.



- ▶ How does revenue vary with the quantity sold?

$$R(Q) = P(Q) \cdot Q$$

with $P(Q)$ the **inverse demand function** (previous example,
 $P(Q) = 8000 - 80 \cdot Q$)

- ▶ From revenue, the **marginal revenue (MR)**:

$$MR = \frac{\Delta R(P, Q)}{\Delta Q} = \frac{R(Q_2) - R(Q_1)}{Q_2 - Q_1}$$

- ▶ Approximated by the derivative (same example):

$$\begin{aligned} MR(20) &= \frac{R(21) - R(20)}{21 - 20} = \frac{P(21) \cdot 21 - P(20) \cdot 20}{1} = 132720 - 128000 = 4720 \\ &\approx R'(20) = 8000 - 160 \cdot 20 = 4800 \end{aligned}$$

- ▶ Selling more requires a lower price: MR adds the price of the extra unit and subtracts the price cut on all the rest...
...hence MR is below the price.

What is feasible

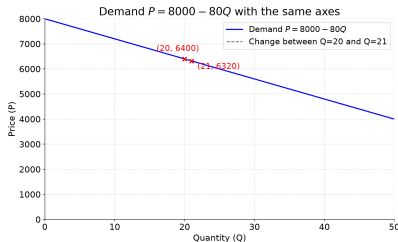
Price elasticity of demand

- ▶ What does marginal revenue depend on?
 - ▶ On how much we have to lower the price
 - ▶ That is, on the sensitivity of quantity to price.
 - ▶ Measured by the **price elasticity of demand**:

$$\varepsilon_P = - \frac{\% \text{change in demand}}{\% \text{change in price}} = - \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = - \frac{P}{Q} \cdot \frac{\Delta Q}{\Delta P}$$

- ▶ At point A, one more unit $\Delta Q = 21 - 20 = 1$:

$$\varepsilon_P = - \frac{\frac{(21-20)}{20}}{\frac{(6320-6400)}{6400}} = 4$$



What is feasible

Elastic demand vs Inelastic demand

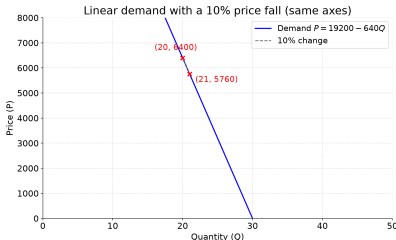
- ▶ $\varepsilon_P > 1$ (previous example): elastic demand.
- ▶ $\varepsilon_P < 1$: inelastic demand.

- ▶ Example:

$$P(Q) = 19200 - 640 \cdot Q$$

$$\varepsilon_P = - \frac{\frac{(21-20)}{20}}{\frac{(5760-6400)}{6400}} = 0.5$$

- ▶ When does a price cut raise revenue?



- ▶ The more elastic the demand, the smaller the price discount in MR.
 - ▶ **Elastic demand** $\varepsilon_P > 1$. Quantity changes more than price (in %), $MR > 0$.
 - ▶ **Inelastic demand** $\varepsilon_P < 1$. Quantity changes less than price (in %), $MR < 0$.
- ▶ In general, what does marginal revenue depend on?

$$MR = P - \frac{\Delta P}{\Delta Q} Q = P \left(1 - \frac{\Delta P}{\Delta Q} \frac{Q}{P} \right) = P \left(1 - \frac{1}{\varepsilon_P} \right)$$

- ▶ EasyJet bet on very elastic markets.
- ▶ High elasticity means large sales.
- ▶ EasyJet started with an investment of less than €6m... and today...

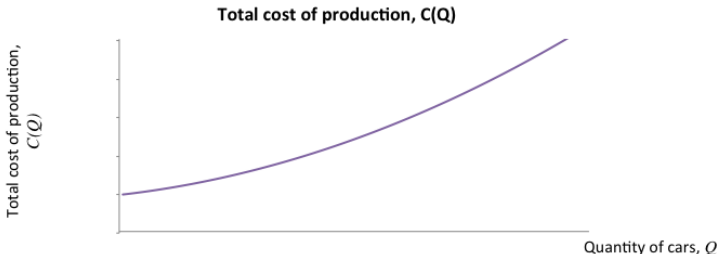
Airline	Revenue	Passengers	Profits
Lufthansa Group	€37.6bn	131.3m	€1.38bn net
IAG (BA, Iberia, Vueling, Aer Lingus, LEVEL)	€32.1bn	122m	€2.73bn net
Air France-KLM	€31.5bn	98m	€1.60bn operating
Turkish Airlines	\$22.7bn	85.2m	\$2.4bn net
Ryanair Group	€13.44bn	183.7m	€1.92bn net
easyJet	€9.309bn	89.7m	€452m net (PBT €610m)
Wizz Air	€5.07bn	62.0m	€365.9m net
Jet2 plc	€6.26bn	17.7m	€529.5m PBT
Pegasus Airlines	€3.1bn	37.5m	€361m net
AEGEAN Airlines	€1.78bn	16.3m	€129.9m net



- ▶ But low prices only pay off with low costs.

4. Costs

- ▶ **Costs**: an essential ingredient of profit.
- ▶ Managers need the **cost function** to set prices and output.
- ▶ **Total cost** $C(Q)$: depends on output Q
- ▶ $C(Q)$ is **increasing** in Q



Classification of costs

- ▶ $C(Q)$ splits into two parts:
 - ▶ **Fixed Costs (FC)**: do not change with Q ;
 - ▶ **Variable Costs (VC)**: change with the quantity produced, Q .
- ▶ Two **concepts of cost**:
 - ▶ **Average cost (AC(Q))**: cost per unit, on average:

$$AC(Q) = \frac{C(Q)}{Q}$$

- ▶ **Marginal cost (MC(Q))**: extra cost of one more unit:

$$MC(Q) = \frac{C(Q_2) - C(Q_1)}{Q_2 - Q_1} = \frac{\Delta C(Q)}{\Delta Q}$$

- ▶ In words:
How much do costs change if we produce a little more?
What was the production cost of the **last unit**?
 - ▶ If $\Delta Q = Q_2 - Q_1 = 1 \Rightarrow MC(Q) = C(Q_2) - C(Q_1) = \Delta C(Q)$
 - ▶ **Compatible with the total cost of class 1** ($C = wL + rK$): only the perspective changes.

What is desirable

Average and marginal costs

- Distinguish **average** from **marginal** cost:

- $AC(Q) = \frac{C(Q)}{Q}$

- $MC(Q) = \frac{\Delta C(Q)}{\Delta Q}$

Quantity (Q)	Total Cost (C)	Average Cost (AC)	Marginal Cost (MC)
...	...	...	...
20	80000	$\frac{80000}{20} = 4000$	—
21	82200	$\frac{82200}{21} = 3914$	$\frac{82200 - 80000}{21 - 20} = 2200$
...	...	...	...
40	136000	3400	—
41	139400	—	—
...	...	...	...
60	216000	3600	4600
61	220600	—	—

- 20 units: average cost €4,000 each.
- One more unit (20 to 21): total cost rises by €2,200.

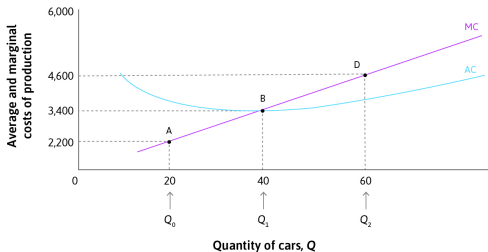
Average and marginal costs

- Example: a university's costs per student.

	Students	MC (\$)	AC (\$)	Total cost (\$)
Undergraduates	2,750	7,259	7,659	21,062,250
	5,500	6,548	7,348	40,414,000
	8,250	5,838	7,038	
	11,000	5,125	6,727	73,997,000
	13,750	4,417	6,417	88,233,750
	16,500	3,706	6,106	100,749,000
	Students	MC (\$)	AC (\$)	Total cost (\$)
Graduates	550	6,541	12,140	6,677,000
	1,100	6,821	9,454	10,339,400
	1,650	7,102	8,672	
	2,200	7,383	8,365	18,403,000
	2,750	7,664	8,249	22,684,750
	3,300	7,945	8,228	27,152,400

Koshal and Koshal (1999)

- ▶ Costs usually satisfy:
 - ▶ Average cost (AC): U-shaped.
 - ▶ Marginal cost (MC): increasing.
 - ▶ AC is decreasing if it is greater than MC, ...
... AC is increasing if it is less than MC.
 - ▶ AC reaches its minimum where it crosses MC



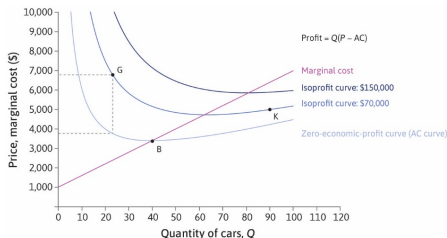
- ▶ Why do these properties make sense?

5. What is desirable: isoprofit

What is desirable

Isoprofit curves

- ▶ **The firm's profit:**
$$\pi(P, Q) = R(P, Q) - C(Q) = P \cdot Q - C(Q)$$
- ▶ Its indifference curves: **combinations of quantity and price with the same profit:**



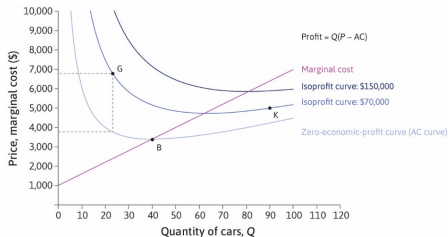
- ▶ **The isoprofit curves**
- ▶ For the profit level k :
$$k = P \cdot Q - C(Q)$$
- ▶ The most obvious is zero profit:
$$0 = P \cdot Q - C(Q)$$
$$P = \frac{C(Q)}{Q} = AC(Q)$$

That is, the average cost curve!!!
- ▶ Higher isoprofit curves, **higher profits.**

What is desirable

Isoprofit curves

- ▶ The isoprofit slope depends on costs:
 - ▶ Isoprofit slope = $\frac{P-MC}{Q} = MRS$



- ▶ If P is constant, increasing Q by 1 unit:
 - ▶ Increases profit by:
 $P - MC$
 - ▶ Thus, **profit per unit** grows: $\frac{P - MC}{Q}$
- ▶ And to keep profit constant, P must decrease by:

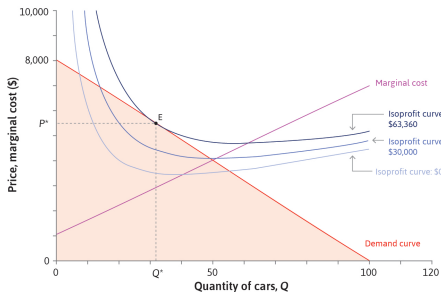
$$\frac{P - MC}{Q}$$

6. Profit maximisation

Profit optimisation

Finding the quantity and price that maximise profit

- Profit is maximised where isoprofit is tangent to demand:



- In this case:

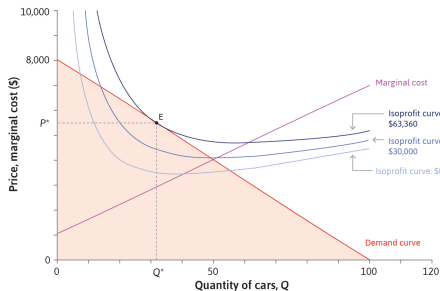
- **Demand curve** = The firm's feasible frontier
[slope = Marginal Rate of Transformation = MRT];
- **Isoprofit curves** = Same profit
[slope = Marginal Rate of Substitution = MRS].

- In equilibrium, $MRT = MRS$

- **Point E** = $(Q^*, P^*) = (32, 5,440)$: last feasible point on the highest isoprofit curve.

Profit optimisation

Finding the quantity and price that maximise profit



- ▶ The slope of demand (MRT) is

$$\frac{\Delta P}{\Delta Q}$$

- ▶ and the slope of the isoprofit curve (MRS) is

$$-\frac{P - AC}{Q}$$

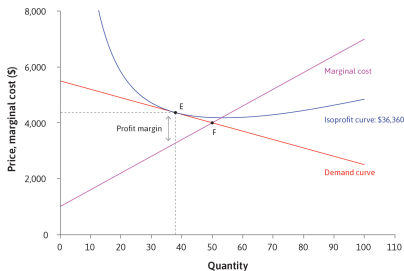
- ▶ Setting them equal we obtain:

$$\frac{P - AC}{P} = \frac{1}{\varepsilon_P}$$

Price elasticity and profits

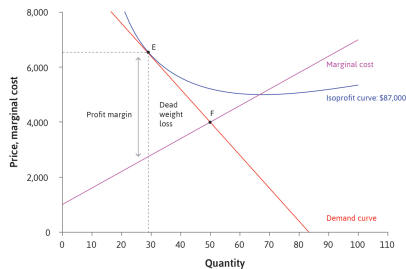
Some markets are more sensitive to price changes than others

- ▶ The firm charges a higher “markup” (margin) the more inelastic demand is.
- ▶ The markup measures the firm’s market power:
 - ▶ How far above marginal cost can it price?



Elastic demand

(Flatter: quantity reacts more to price)



Inelastic demand

(Steeper: quantity reacts less to price)

- ▶ What determines **market power** and, more generally, the elasticity of demand?



- ▶ **Very similar products have different prices;**
- ▶ This is due to **product differentiation**.



(1801-1877)

- ▶ **Antoine Cournot** modelled imperfect competition observing the bottled water market.

- ▶ **Product differentiation** can come from:
 - ▶ **Innovation:** More features (smartphone vs mobile phone, hybrid vs conventional cars) or entirely new products.
 - ▶ **Advertising:** brand loyalty gives large margins (Apple versus Huawei)
 - ▶ Both shift the firm's demand curve.

Section by section (Micro, Unit 7)

1. **Introduction** → a firm chooses price and quantity; competing on low prices can make it the market leader.
2. **The firm's problem** → $\pi = R(P, Q) - C(Q)$: first what is feasible, then what is desirable.
3. **What is feasible: demand** → the demand curve is the firm's frontier; selling one more unit requires a lower price, so $MR < P$.
4. **Costs** → AC is U-shaped, MC increasing, and AC reaches its minimum where it crosses MC .
5. **What is desirable: isoprofit** → isoprofit curves are the firm's indifference curves; the zero-profit one is the AC curve.
6. **Profit maximisation** → tangency between demand and isoprofit: $MRS = MRT$, equivalently $MR = MC$.
7. **Elasticities and markups** → $\frac{P-MC}{P} = \frac{1}{\varepsilon_P}$: the more inelastic the demand, the larger the markup and the market power.

Key concepts: demand curve, inverse demand, marginal revenue, price elasticity, elastic and inelastic demand, fixed and variable costs, average and marginal cost, isoprofit curve, zero-economic-profit curve, $MRS=MRT$, markup, market power, product differentiation.